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SECTOR & TRADE: ICT & Multimedia, Networking and Internet Technologies
RQF LEVEL 3 | ACADEMIC YEAR 2025-2026 | 8rd December 2025
MODULE TITLE APPLY C PROGRAMMING FUNDAMENTALS
DURATION 3 Hours
MAXIMUM MARKS 100 Marks
Trainer MPAYIMANA CYIZA Landry
Trainee **……**

INSTRUCTIONS TO STUDENTS

- This paper contains **Section A** and **Section B**.
- **All questions in Section A are compulsory (70 Marks).**
- **Attempt ANY TWO (2) questions in Section B (30 Marks).**
- Write clearly; unclear answers score zero.

SECTION A (70 Marks) 14 Questions × 5 Marks Each — ALL COMPULSORY

Q1. A programming language that uses binary code made of 0s and 1s is called _____.

Q2. Which of the following is a high-level programming language?
A. Machine language
B. Assembly language
C. C
D. Binary language

Q3. Match the language type to its description:

1. Assembly language
2. High-level language
3. Machine language

A. Easy for humans to read and write
B. Uses symbolic names (mnemonics)
C. Uses 0s and 1s

Q4. High-level programming languages are designed to be easy for humans to understand.

Q5. Which symbol is used in a flowchart to represent START/END?
A. Diamond
B. Oval
C. Rectangle
D. Parallelogram

Q6. A step-by-step procedure used to solve a problem is called an _____.

Q7. Which of the following is an example of a C programming IDE?
A. Excel
B. Dev-C++
C. WordPad
D. Python IDE

Q8. An IDE contains a source code editor, build tools, and a debugger.

Q9. Which compiler comes with Dev-C++ during installation?
A. HTML Compiler
B. MinGW/GCC
C. Java Engine
D. Pascal Compiler

Q10. Python is used to develop network _____ and simulate telecommunication networks.

Q11. Which IDE supports many languages and is accessed through the Internet?
A. Eclipse
B. Online IDE
C. Code::Blocks
D. Turbo C

Q12. Assembly language uses mnemonics to represent machine instructions.

Q13. DSP stands for Digital _____ Processing.

Q14. During Dev-C++ installation, the user must first click:
A. Remove
B. I Agree
C. Delete
D. Cancel

SECTION B (30 Marks) Answer ANY TWO Questions Only Each Question = 15 Marks

Q15.

A. Describe the difference between low-level and high-level programming languages.
B. Give two examples of high-level programming languages.
C. State one advantage of using high-level languages.
(5 + 5 + 5 marks)

Q16.

A. Explain the purpose of a flowchart in programming.
B. Identify any two flowchart symbols and state their meaning.
C. Explain why a flowchart must follow a logical sequence.
(5 + 5 + 5 marks)

Q17.

A. Define an Integrated Development Environment (IDE).
B. Mention any two types of C programming IDEs.
C. Describe any two steps involved during the installation of Dev-C++.
(5 + 5 + 5 marks)

Q18.

Discuss how you can run your first C programming program.
Include all necessary steps, starting from opening an IDE /15 Marks

printf(“GOOD LUCK!!!!!!!!!!!!!!!”) ;